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A Genomic Perspective on Species Delimitation

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Abstract

Genomic species delimitation is transforming how we understand and define species by enabling a process-oriented and efficient approach to identifying species boundaries. This review outlines the two key steps in genomic species delimitation: (*a*) discovering species-level units and (*b*) assessing their validity. Validity can be evaluated by a diversity of approaches, including applying the multispecies coalescent to delineate the population–species boundary and using estimated gene flow as a proxy for reproductive isolation. We illustrate the utility of these methods across the tree of life through a comprehensive review of published articles and case studies on birds, siphonophores, and bacteria. Despite the many benefits of genomic species delimitation, challenges remain. In particular, genomic divergence does not always accurately reflect ecological divergence and reproductive barriers, and genome heterogeneity can complicate the overall understanding of genetic divergence. We discuss these challenges and potential solutions.

INTRODUCTION

Whether walking through urban neighborhoods or hiking through natural areas, an unbelievable diversity of living forms surrounds us, from insects and birds to microscopic bacteria and protozoans. The basic units of this biodiversity are species, unique evolutionary lineages in the tree of life. Despite the central importance of species, biologists cannot agree on the number of species on Earth, with estimates ranging from 2 million to 100 million (Mora et al. 2011). Yet, knowing the number of species and their boundaries is crucial across biology—from interpreting molecular biological processes to identifying and surveilling pathogenic species to conserving landscapes. In particular, studies of biodiversity from population- to broad-level scales rely on having an accurate taxonomy in order to understand how species form and are maintained.

WHAT IS GENOMIC SPECIES DELIMITATION?

One way to define species boundaries is through genomic species delimitation, or the use of genomic data to circumscribe the number of species in a clade and the boundaries among them (**Figure 1**). Before we can delimit species, we require a species concept. Although there are seemingly as many species concepts as there are biologists who study speciation, in this review, we define a species as a cohesive metapopulation that remains distinct because of interbreeding and/or ecological forces (Maddison & Whitton 2023). By this definition, evaluating whether two sympatric lineages are distinct species is relatively straightforward because their reproductive and ecological interactions can be directly observed. However, allopatric lineages pose a greater

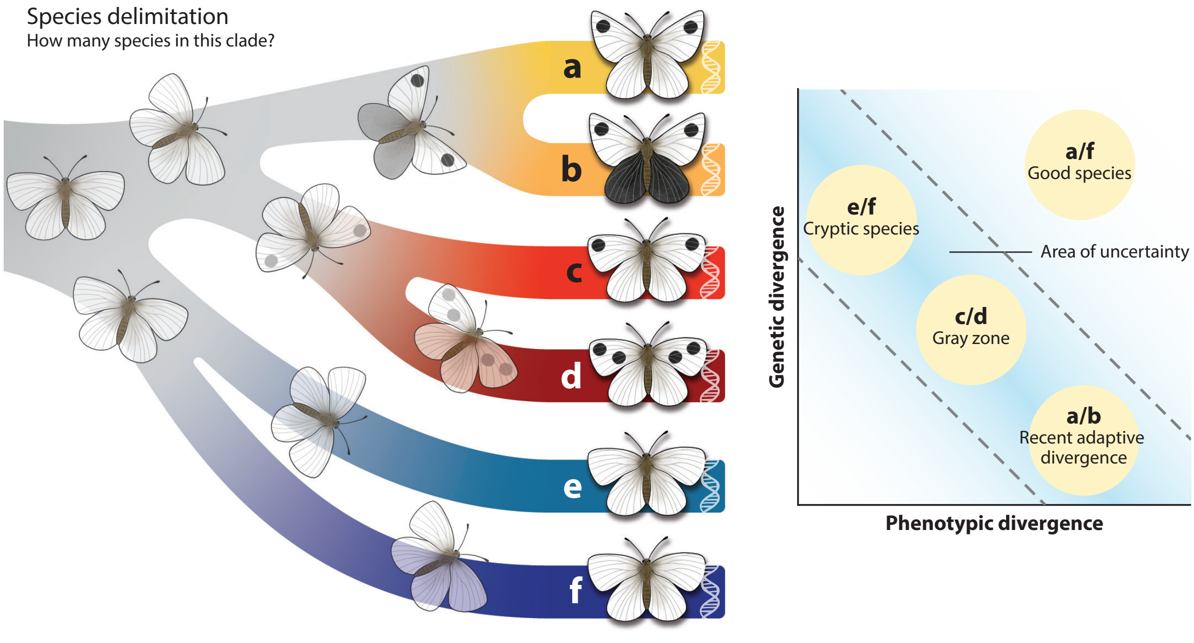


Figure 1

The goal of species delimitation is to define the number of species in a clade and the boundaries between them. Here we focus on how genetic and phenotypic divergence can be used to define species. When species are markedly different genetically and phenotypically (e.g., butterflies *a* vs. *f*), then they are clearly good species. However, when levels of genetic and phenotypic divergence are uncorrelated (e.g., butterflies *e* vs. *f* and *a* vs. *b*) or when divergence is intermediate (e.g., butterflies *c* vs. *d*), species status can be ambiguous.

challenge, because their independence must be evaluated using proxies rather than by directly studying their interactions in nature.

Historically, systematists have used morphological data as a proxy to evaluate the distinctiveness of lineages. Under the assumption of random mating and polygenic inheritance, morphological differences between populations reflect the influence of distinct evolutionary forces that prevent the populations from merging. Therefore, such morphological differences can strongly indicate species distinctiveness (Cadena et al. 2018, Dobzhansky 1937).

With the advent of karyotyping, allozyme data, and Sanger sequencing, biologists gained another axis along which to evaluate lineage distinctiveness: genetic divergence. These data revealed that genetic and phenotypic perspectives on lineage divergence can disagree, such as in the case of recent adaptive radiations or cryptic speciation (**Figure 1**). The development of high-throughput sequencing in the early 2000s further revolutionized this field, enabling a transition from genetic to genomic perspectives on species divergence. Genetic and genomic datasets differ by the number of independent loci sampled; while genetic datasets have traditionally sampled the organellar genome (mitochondrial or chloroplast DNA) or a handful of nuclear loci, genomic datasets span hundreds or thousands of loci to reveal genome-wide patterns of differentiation. Genomic data have thus provided an unprecedented window into genetic divergence and a powerful new tool for species delimitation. In this review, we explore the utility and practice of genomic species delimitation, discuss advantages and disadvantages, and suggest future directions and possibilities for the field.

GENOMIC SPECIES DELIMITATION IN PRACTICE

Data Types

Supplemental Material >

Genomic species delimitation typically relies on data collected using one of two primary methods (**Supplemental Table 1**) (da Fonseca et al. 2016). The first uses reduced-representation approaches, in which researchers subset and sequence a homologous portion of the genome across their samples. While these approaches do not require a reference genome, a reference genome can ease analyses by making homology identification more accurate. One popular reduced-representation approach is restriction site–associated DNA sequencing (RADseq) (Puritz et al. 2014), in which the genome is digested with restriction enzymes, and then fragments are selected by size and sequenced. This method is cost-effective and requires minimal initial set-up costs, but identifying homologous loci can become challenging as targeted populations become more genetically divergent (Leaché & Oaks 2017). Another reduced-representation method is targeted sequence capture, in which any portion of the genome is baited with probes and then sequenced. Probes can be designed to capture (among others) exonic regions, restriction-aided digest loci, or highly conserved loci. This approach requires initial investment in both probe design and production, but it can be used on a diversity of samples (including museum- or herbarium-preserved specimens) and often returns more complete data matrices compared to other approaches. A final reduced-representation method is to sequence the transcriptome, i.e., the expressed portion of the genome. Because this method uses RNA, it requires fresh, well-preserved tissues, often necessitating specialized logistics for collection and preservation (e.g., liquid nitrogen or freezers). Despite these logistical challenges, transcriptome data have utility beyond species-delimitation studies because they enable researchers to explore functional genomic questions as well.

The second general approach is whole-genome resequencing, in which the entire genome is sequenced across all samples. This approach typically requires a reference genome or a metagenomics approach, though genome skimming approaches—in which researchers assemble genomic regions de novo—provide a genome-free pathway (Straub et al. 2012). In either scenario, this approach remains most affordable and tractable for species with small genomes. While sequencing the genome might seem excessive for species delimitation, genome sequences can allow a richer understanding of species and their biology. For example, genomes can illuminate the evolution of structural variants, enable functional genomics studies, and facilitate the characterization of patterns of selection and introgression across the genome.

Genomic Species-Delimitation Approaches

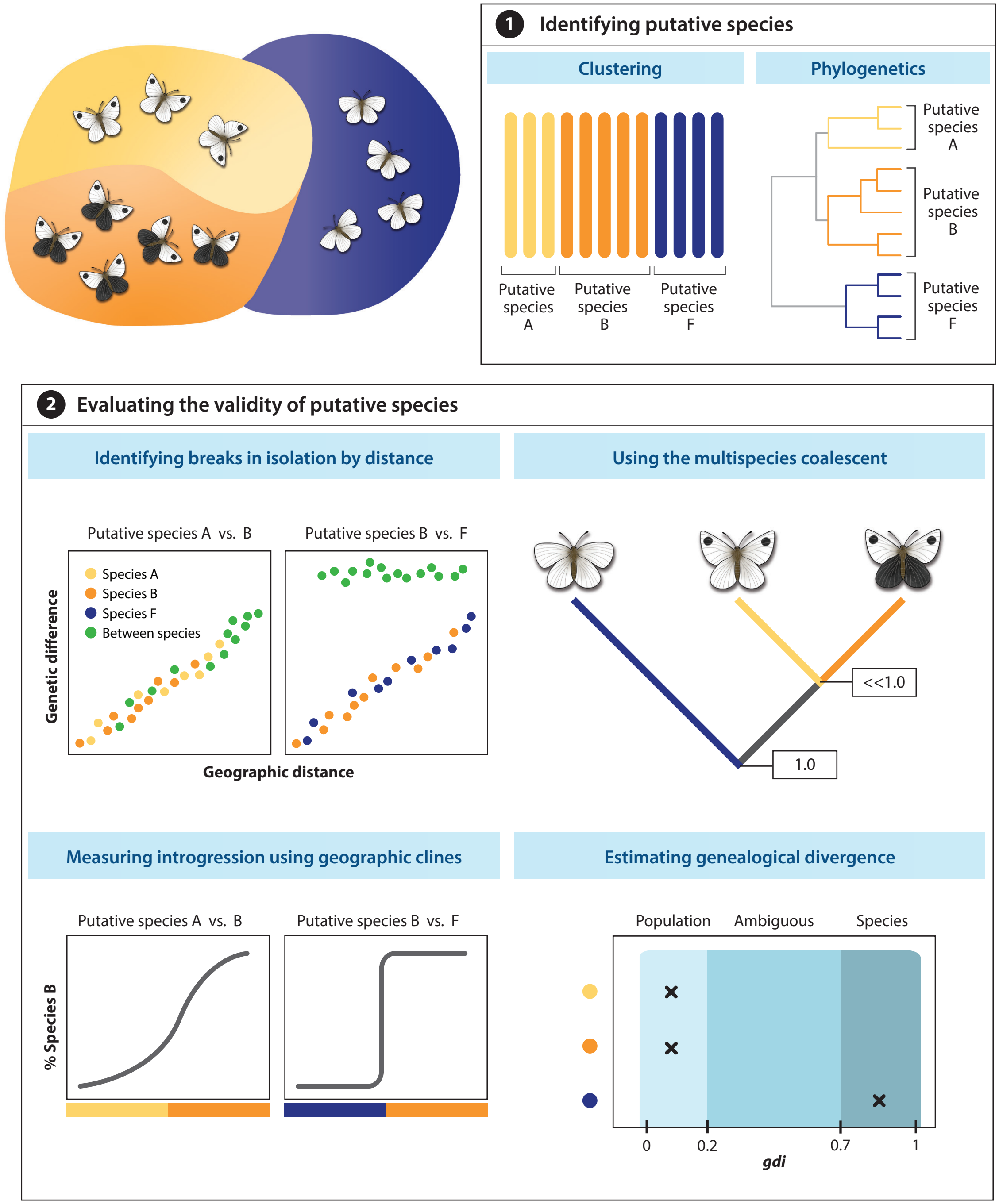
Species delimitation typically proceeds via two steps (**Figure 2**). First, researchers identify the putative species-level units (discovery), and second, they evaluate the validity of these units (validation) (Carstens et al. 2013). In the next two sections, we outline methods and approaches used in each of these steps (**Table 1**, **Supplemental Table 2**), including their strengths and weaknesses.

Supplemental Material >

Step 1: Identifying species-level units. At the start of a species-delimitation study, researchers often assume predefined species designations based on traditional taxonomy, genetic datasets, or morphological groupings. However, more commonly, researchers use newly collected data to propose species hypotheses, by inferring either phylogenetic trees or population structure.

After building an individual-level phylogenetic tree, species-level units can be defined by identifying monophyletic groups. This approach is efficient but limited: Good species are often not monophyletic, whether due to incomplete lineage sorting, introgression, or other sources of genealogical discordance. Additionally, phylogenetic trees are inherently hierarchical, allowing monophyletic groups to be identified at various phylogenetic depths and making it unclear where the appropriate threshold for species delimitation lies. One algorithmic solution identifies this threshold by determining where phylogenetic branching patterns transition from reflecting within-species coalescent patterns to between-species divergence (Fujisawa & Barraclough 2013). While methods like the Generalized Mixed Yule Coalescent (GMYC) approach were designed for single-locus phylogenies, they have been extended to phylogenomic tree datasets (Jacobs et al. 2021).

Additionally, genomic data can identify genotypic clusters, which are then treated as candidate species-level units. One approach involves visualizing individuals in genotypic space using dimensionality reduction techniques, such as principal component analysis (PCA) (Price et al. 2006). Individuals' proximity to each other reflects their level of genetic divergence, and individuals forming genetically cohesive units typically cluster in genotypic space. Although descriptive, approaches like PCA allow researchers to quickly identify putative species-level clusters. More formal clustering approaches use genotypic data to group individuals into genetic populations or demes [e.g., as implemented in the popular program STRUCTURE (Pritchard et al. 2000)]. Genetic clustering methods identify the characteristics of the genotypic data that suggest individuals are interbreeding—for example, they identify the genetic populations that maximize Hardy–Weinberg equilibrium and minimize linkage disequilibrium within a set of individuals. Alternatively, genotypic data can be transformed into distances based on allele frequencies and sharing, and clusters can be identified via Gaussian mixture modeling (Hausdorf & Hennig 2010). While clustering methods are powerful, they require the user to determine the number of genetic populations a priori. When users want to test the number of species (or genetic populations) in a set of individuals, they must determine which model (i.e., the number of genetic populations and the assignment of individuals to genetic populations) describes the data best. Further, these approaches do not fully account for the role of geography in structuring populations (but see Bradburd et al. 2018).



(*Caption for **Figure 2** appears on following page*)

Figure 2 (*Figure appears on preceding page*)

Approaches to genomic species delimitation. After sampling broadly across the geographic ranges of putative lineages, species delimitation typically occurs in two steps. In Step ①, genomic data are used to define the putative species, typically through either phylogenetic or genotypic clustering approaches. In Step ②, genomic data can be used to test the validity of these species, through methods such as identifying breaks in isolation by distance between individuals across putative species, calculating the posterior probability of alternative species delimitations using multispecies coalescent approaches, measuring levels of introgression by inferring clines across parapatric boundaries, and estimating the genealogical divergence index (*gdi*). Here, all four methods show concordant results: Genomic data support the split separating putative species A and B from F but not that between putative species A and B. Individual researchers would need to decide how to reconcile the discordant phenotypic and genetic divergence of putative species A and B.

Step 2: Evaluating the validity of species-level units. After identifying putative species-level units, researchers typically evaluate their validity by determining whether these species exhibit evidence for reproductive isolation or genetic, ecological, or morphological distinctiveness (Padial et al. 2010). Here, we focus on three major classes of methods that can help test the validity of species-level units: non-tree-based methods, tree-based methods, and gene-flow methods.

Non-tree-based methods. There are three commonly used methods to validate candidate species that do not require a phylogenetic framework. The first looks for discontinuities in isolation by distance (Chambers et al. 2025, Good & Wake 1992, Hausdorf 2025). In many species, because gene flow is more likely between geographically proximate individuals, genetic differentiation increases with geographic distance, creating a continuous pattern of isolation by distance. However, for individuals from different species, they are further isolated by reproductive barriers, leading to discontinuities in isolation by distance (Hausdorf & Hennig 2020). Identifying such discontinuities can help identify species-level units. This method explicitly accounts for geography but requires extensive sampling, which can limit its feasibility.

The second approach calculates metrics of genetic divergence (i.e., F_{ST}, d_{xy}) among putative species and then assigns species-level status to those above a certain threshold. This approach is essentially a more data-rich version of DNA barcoding (Moritz & Cicero 2004). As with DNA barcoding, identifying an appropriate threshold to demarcate the population–species boundary can seem arbitrary, though data from species known not to interbreed (e.g., those in sympatry) (Mayr 1969) or barcoding gaps between intra- and interspecific genetic divergence (Čandek & Kuntner 2015, Hart et al. 2025) can help approximate an appropriate level.

Third, machine-learning methods are particularly notable for their flexibility and efficiency (Karbstein et al. 2024). In supervised learning, researchers train the algorithm using data from known species (e.g., training data); by identifying the features of known species, likely species can then be circumscribed using new data. Unsupervised learning works similarly but does not require training data. Machine-learning approaches can use genomic data alone, or they can integrate across genomic and phenotypic data to identify species-level clusters (Pyron 2023).

Tree-based methods. Tree-based methods define species boundaries based on phylogenetic relationships or genetic divergence values derived from the phylogeny. Most tree-based methods rely on a multispecies coalescent (MSC) framework (Rannala & Yang 2003). At their core, species-delimitation approaches based on the MSC extend the coalescent process of gene genealogies within a population—dependent on demographic parameters such as population size and generation time—to multiple populations connected by a species tree (Fujita et al. 2012). These methods provide a statistical comparison of alternative species-delimitation models by collapsing or splitting nodes of the species tree (Yang & Rannala 2010) or by estimating the marginal likelihoods for alternative models and comparing them with Bayes factors (Leaché et al. 2014) (**Figure 2**). The coalescent estimators (e.g., divergence times, population sizes, introgression rates) can also directly inform species status. For example, the genealogical divergence index (*gdi*) is calculated

Table 1 Genomic methods for identifying putative species-level units and evaluating their validity[a]

Stage in species delimitation	General class of methods	Method	Exemplar program(s)
Step 1: Identifying putative species-level units	Genotypic methods	Using principal component analysis to ordinate genotypic data and visually identify clusters (i.e., putative species)	adegenet, PLINK
		Applying model-based or Gaussian mixture model clustering methods to genotypic data to identify genetic populations (i.e., putative species)	ADMIXTURE, sNMF, prabclus
	Phylogenetic methods	Visually identifying putative species as monophyletic clades in phylogenetic trees	RAxML, IQ-TREE
		Automatically identifying putative species as monophyletic clades in a phylogeny (while not designed for genomic data, the method has been applied in this way)	bPTP, GMYC
Step 2: Testing the validity of species-level units	Non-tree-based methods	Evaluating whether putative species exhibit a discontinuity in isolation by distance	prabclus
		Determining how putative species level of genomic divergence compares to a given population–species threshold	FastANI
		Applying machine learning to species delimitation	CLADES
	Tree-based methods	Applying the multispecies coalescent to determine when coalescent patterns switch from within to between species	BPP, BFD, hhsd
		Applying the protracted speciation model to determine the likelihood that lineages evolve into good species	DELINEATE
		Jointly inferring demographic history and species delimitation	PHRAPL, delimitR
	Methods using gene flow as a proxy for reproductive isolation	Inferring introgression during species divergence from topological and/or branch length variation	Dsuite, HyDe
		Inferring levels of gene flow during species divergence through demographic modeling	G-PhoCS, dadi
		Identifying introgression by inferring geographic or genomic cline width at parapatric boundaries between putative species	HZAR, bgc
		Identifying introgression by looking for admixed individuals (often at parapatric boundaries between putative species)	ADMIXTURE, sNMF
		Identifying introgression by inferring phylogenetic networks	SNaQ, PhyloNet

[a]A more exhaustive list of methods (including relevant citations) is available in **Supplemental Table 2**.

using the population sizes and divergence times estimated from the MSC (with or without migration) to conduct species delimitation (Jackson et al. 2017, Leaché et al. 2019). The *gdi* measures the probability that two alleles sampled from a species will coalesce with each other before they coalesce with an allele from another species. A useful characteristic of the *gdi* is that it provides a

Supplemental Material >

relative measure for comparing populations and species on a scale that ranges from 0 (panmictic population) to 1 (species-level divergence) (**Figure 2**).

Tree-based approaches have some technical limitations. Some implementations require the user to have a phylogenetic hypothesis for the sampled individuals (a guide tree) (Yang & Rannala 2010), and some cannot scale to genome data (but see Jackson et al. 2017, Rabiee & Mirarab 2020). Yet, they have been shown to return statistically consistent results across a wide range of divergence times, population sizes, and sampling schemes (Leaché et al. 2019, Luo et al. 2018, Zhang et al. 2011). However, this statistical robustness does not imply that these units conform to any species concept. Population structure within species is ubiquitous, and if gene flow is limited, then the MSC often delimits within-species populations as species (Chambers et al. 2025, Sukumaran & Knowles 2017). Perhaps it is unsurprising then that MSC-defined species sometimes show little to no evidence of reproductive isolation, suggesting they are not acting as biological species (Campillo et al. 2020, Singhal et al. 2018).

Process-based methods might resolve some of these issues. One such approach builds upon the protracted speciation model (Etienne et al. 2014), which recognizes that speciation is not an instantaneous process—rather, lineages must persist for some waiting time before they become good species (Sukumaran et al. 2021). This approach introduces a parameter that captures the likelihood that an incipient lineage graduates to being a good species (the speciation completion rate). Estimating this parameter requires the user to provide known cases where sampled lineages are distinct species, which then helps constrain the estimation of the completion rate parameter. As yet, this approach has not been tested with many empirical datasets, but it provides a promising path forward. Another approach involves evaluating model fit across demographic models that represent multiple, competing species delimitations (Smith & Carstens 2020). For example, with four putative species, scenarios can be specified ranging from one to four species, with further customization for population size changes, gene flow events, and different topologies. Which scenario best fits the data is then determined, thus identifying both the most likely number of species and their demographic history.

Inferring gene flow as a proxy for reproductive isolation. A final class of approaches uses genomic data to infer levels of hybridization and introgression, which are used as a proxy for the extent of reproductive isolation. If putative species-level units show evidence of limited to no gene flow, then they are assumed to be isolated by reproductive barriers and are ascribed species status.

A variety of methods exist for identifying gene-flow events, and here we outline some common ones used in species-delimitation studies. One popular approach compares topological or branch-length variation across evolutionary trees to determine where introgression has occurred (Hibbins & Hahn 2022). An example is the D statistic (also known as the ABBA–BABA test), which relies on the assumption that populations with an excess of shared variation have exchanged genes in the past (Durand et al. 2011). Additionally, coalescent-based methods can infer the demographic history of gene flow and introgression in a group (Flouri et al. 2023, Jackson et al. 2017, Smith & Carstens 2020). Typically, researchers first evaluate which of several demographic models best fit the data, often comparing models in which populations are completely isolated with ones in which they exchange genes. Then, parameters for the best-fitting model are estimated, allowing researchers to infer the approximate level of gene flow between species. Another approach uses genomic data to quantify admixture at contact zones or hybrid zones between putative lineages. The width of geographic clines between hybridizing putative species can indicate the strength of reproductive isolation, with steeper clines indicating greater isolation (**Figure 2**) (Singhal et al. 2018). Because measuring clines requires dense geographic sampling, a more tractable method is to look for evidence of admixed individuals in regions of sympatry or parapatry between putative species. Such individuals can have mixed nuclear ancestry and/or mismatched nuclear and organelle genomes. A final approach uses phylogenetic networks to infer instances of introgression and hybridization among species-level tips (Hibbins & Hahn 2022, Huson & Bryant 2006).

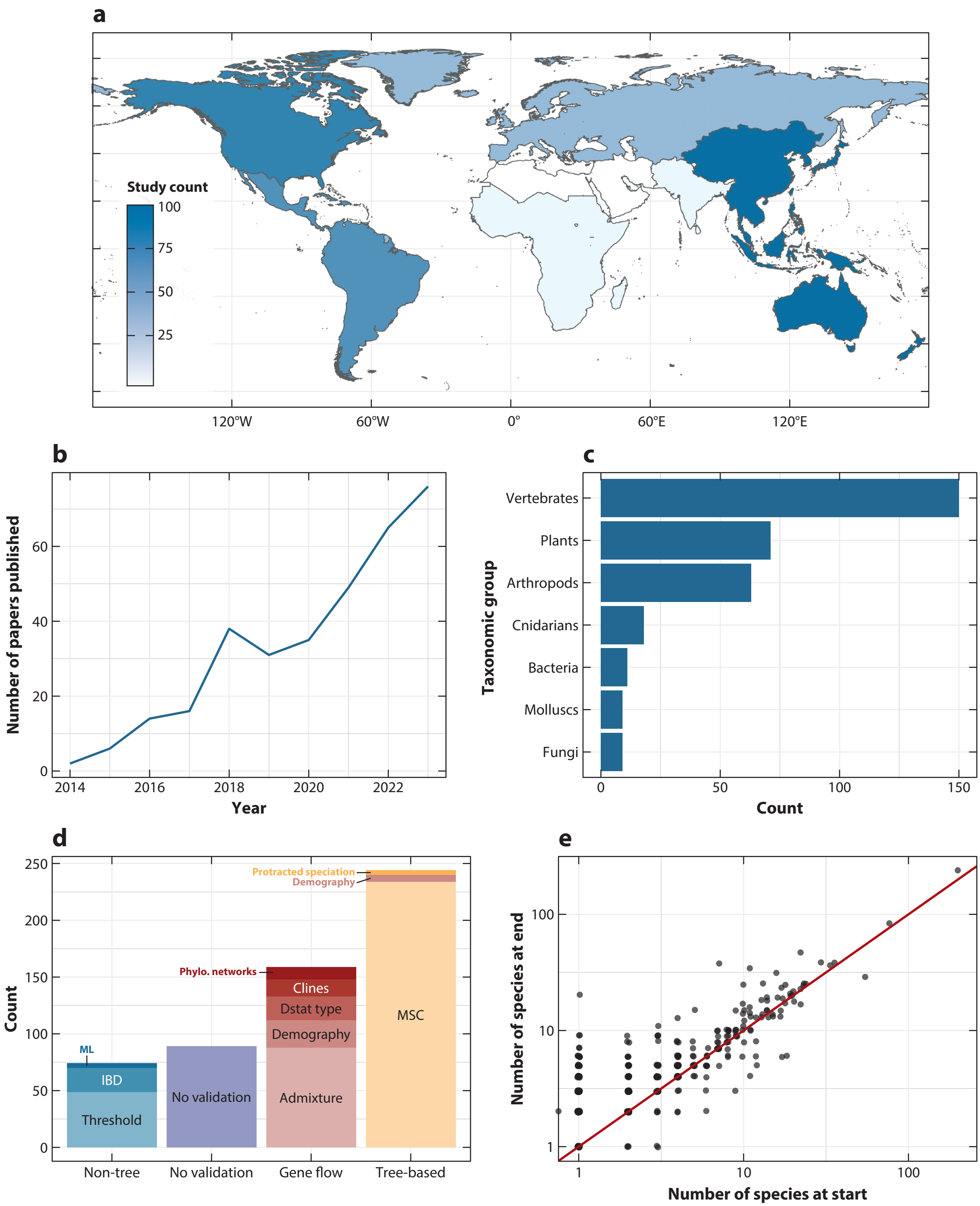
This general class of gene-flow methods shares three, interlinked challenges. First, hybridization and introgression are rampant across the tree of life (Dagilis et al. 2022), even across lineages that otherwise appear to be quite ecologically distinct and mostly reproductively isolated. Thus, how much hybridization is too much? Should species show complete isolation? Or can a modest exchange of individuals be tolerated, or even expected? If lineages are able to remain phenotypically distinct despite substantial genetic exchange, should they still be considered distinct species (Cadena & Zapata 2021)? Second, estimating gene flow accurately is challenging (Martin et al. 2015, Roux et al. 2016), so relying on gene-flow levels to determine species status could be error-prone. Third, historical gene-flow patterns are not necessarily a good proxy for contemporary reproductive isolation. For example, because dating introgression events is challenging, apparent evidence for introgression in modern populations could result from historical gene flow between populations that are reproductively isolated in the present day. Or, if putative species have infrequent opportunities to mate—i.e., they have limited geographic range overlap or occur at low densities in contact zones—then gene flow would be low even if reproductive isolation is incomplete. Thus, gene flow estimates can be a misleading indicator of current levels of reproductive isolation.

A Review of Genomic Species Delimitation

To understand how researchers use genomic data to delimit species, we searched "species delimit*" in Web of Science in October 2024 for papers published from 2013 to 2023, downloading the full records of the 4,896 hits (see also Bergman et al. 2025). We then identified which papers putatively used genomic data by searching for a match to any of the following key phrases—i.e., "genomic", "RAD", "SNP", "target capture"—in the title and abstract of these papers, resulting in 343 matches. We then reviewed all these genomic species-delimitation studies, cataloging the taxonomic group, world region, genomic markers and methods used, whether nongenomic data were used, and the final delimitation results (how many species were identified relative to the starting taxonomy and whether the taxonomy was formally updated).

While applying genomic methods to species delimitation is growing increasingly popular (**Figure 3**), ultimately only a small proportion (7%) of all species-delimitation studies ($n = 4{,}896$) employed genomic data. (Genetic data were much more common; a random sample of 5% of all 4,896 papers suggests ~90% of papers used genetic data to delimit species.) The vast majority of genomic studies ($n = 343$) used either restriction-aided approaches (61%) or target-capture approaches (18%); whole-genome resequencing remains rarer (11%). Genomic species delimitation is much more common in some taxonomic groups and geographic regions than others (**Figure 3**). For example, 44% of reviewed studies focused on vertebrates, even though vertebrates are thought to be less than 5% of the total species diversity on Earth, and only 5% of studies featured species from sub-Saharan Africa, even though species richness is particularly high in equatorial regions of Africa (Freeman & Pennell 2021, Linck & Cadena 2024).

While many studies explicitly set out to test taxonomic hypotheses outlined by previous publications or datasets, nearly all (94%) used phylogenetic or genotypic methods to identify putative species at the start of the study (**Figure 2**, **Table 1**). The majority of studies (73%) then conducted one or more genome-based analyses to test the validity of these species. The most common approach applied was the MSC model, followed by assessing evidence of admixture using genotypic clustering methods. Rarely was the presence of admixture used to collapse species; more often, researchers argued for the validity of delimited units due to the absence of admixture.



(*Caption for* ***Figure 3*** *appears on following page*)

Figure 3 (*Figure appears on preceding page*)

Summary of papers on genomic species delimitation published from 2013 to 2023 ($n = 343$; no papers were published in 2013). Studies of genomic species delimitation (*a*) span the globe but are not evenly distributed, (*b*) have increased steadily through time, and (*c*) are most common in vertebrates. (*d*) Most studies (73%) apply one or more methods to validate putative species, the most popular of which involve the MSC. (*e*) While most studies using genomic species delimitation identify more species-level taxa than at the start, 36% return fewer or the same number of species. For panels *c* and *d*, taxonomic groups and methods used in two or fewer studies are not depicted. Abbreviations: Dstat, D statistic; IBD, isolation by distance; ML, machine learning; MSC, multispecies coalescent; Phylo, phylogenetic.

Despite concerns that genetic approaches are overly zealous in splitting species, 36% of studies identified either the same number or fewer species than the initial taxonomy. In those cases where genomic species delimitation increased the number of species, the median increase in species number was twofold. In many cases, the genomic species delimitation aligned closely with species delimitations inferred from smaller mitochondrial or multilocus datasets, suggesting that smaller genetic datasets can delimit species as accurately as genomic datasets. Many studies refrained from revising taxonomy, commenting that they would enact taxonomic change after they further validated findings through additional sampling or phenotypic data. In the end, only 36% of the studies proposed formal taxonomic changes; notably, studies that revised taxonomy were 2.2 times more likely to include phenotypic data.

Case Studies

Redpoll finches. Species limits in vertebrates are among the best-characterized of any clade, yet they remain subject to revision as new datasets, particularly genomic datasets, become available. One notable example involves the redpoll finches (genus *Acanthis*), a complex of small birds with broad, circumpolar Arctic distributions. While the number of species in *Acanthis* had been unstable across taxonomic treatments, for many years, the leading hypothesis suggested the genus contained three species, defined primarily by differences in plumage and beak morphology.

However, mitochondrial and multilocus datasets found no evidence to support the distinctiveness of these three species (Marthinsen et al. 2008). A genomic RADseq dataset corroborated these findings, showing that the three species were nearly genetically identical and behaved like one species under the MSC model (Mason & Taylor 2015). Yet, transcriptomic data revealed distinct patterns of gene expression among sympatric species that correlated with differences in plumage and beak morphology. Whole-genome resequencing data identified the likely genomic architecture for these differences: a >55-megabase inversion encompassing genes linked to plumage coloration and beak morphology (Funk et al. 2021). Because recombination is reduced in inverted regions, inversions can act like supergenes. As might be expected under such a model, genetic divergence between the taxonomic species was elevated within the inversion. Outside of this region, however, the taxonomic species exhibited high gene flow and minimal divergence. Based on these genomic data, and evidence for continuous variation in genotype and phenotype across geography, researchers proposed collapsing the three species into one, a change later accepted by a committee overseeing the taxonomy of North American birds (Chesser et al. 2024). Redpoll finches exemplify how genomic data can both characterize the species boundaries among species and infer the likely processes that gave rise to the phenotypic diversity seen today.

Man-o'-war siphonophores. The Portuguese man-o'-war (genus *Physalia*) is a widespread siphonophore found in the pelagic zones of oceans worldwide. It floats on the water's surface using a large gas-filled bladder, topped with a sail that allows it to drift with surface winds. This long-distance dispersal was thought to maintain genetic cohesion across its extensive geographic range, and until recently, *Physalia physalis* was considered the only species in the genus. This assumption went unchallenged until 2012, when mitochondrial DNA from individuals in a limited part of the range of *Physalia* (Australia and New Zealand) revealed extensive cryptic diversity (Pontin & Cruickshank 2012). However, the broader implications of this finding remained unclear until systematists conducted the first global genetic and morphological analysis of the species (Church et al. 2025). Whole-genome resequencing of over 100 individuals from across *Physalia*'s global range revealed the existence of five distinct and monophyletic lineages, each highly divergent from the others. Minimal introgression, even among sympatric lineages, suggested the existence of reproductive barriers. A community science dataset containing more than 10,000 photographs of *Physalia* complemented the genomic results, demonstrating that these genomic-based lineages correspond to distinct, morphologically diagnosable units. Further analysis revealed that these morphological groupings aligned with three historical species names within the genus, which had been synonymized over the past century. Based on their findings, the researchers reinstated these species names, increasing the recognized number of *Physalia* species to four. The fifth lineage, pending further data, remains unnamed for now. *Physalia* illustrates how the species boundaries for even widely known species can be inaccurate and how many species—particularly those that are wide-ranging—would likely benefit from a thorough, range-wide appraisal of genetic diversity and structure.

Bacteria. Species delimitation in bacterial species typically proceeds very differently than it does in other groups (Shapiro & Polz 2015). Bacteria are exceptionally diverse, with most species unknown to science, and because many species cannot be easily cultured, collecting phenotypic data is challenging. Further, because bacterial species are thought to reproduce mostly asexually, the forces that maintain cohesion in bacterial species are unclear. Because of these challenges, bacterial species are often diagnosed on a simple threshold of genetic divergence, whether based on barcoding genes or full genome sequences.

Arevalo et al. (2019) offer a different approach, motivated by the observation that bacteria often engage in recombinogenic exchange of genetic material. Designed to be applied to bacteria that are genetically similar, their approach calculates a measure of genome homogenization between all pairwise genome comparisons. Genome homogenization reflects how similar two genomes are due to recent gene flow, which is then used to build a gene flow network. Clusters in the network are distinct populations, which share many of the same properties of animal or plant species—e.g., there is free interbreeding within a cluster and minimal interbreeding between clusters. Applying this delimitation approach to *Ruminococcus gnavus*, a common bacteria found in the human gut, the researchers identified three lineages. They then analyzed these lineages' genomes to understand the unique genetic and biological attributes of each lineage. Each lineage had specific variants that predicted whether it was more likely to be found in healthy people or those with inflammatory bowel disease, and each had experienced specific selective sweeps that might reflect its adaptation to these distinct ecosystems. *Ruminococcus* provides an exciting perspective on how gene flow and adaptive potential can be assessed across the entire genome. In doing so, this approach both diagnoses species and provides insights into species-specific biology and ecology.

ADVANTAGES OF GENOMIC SPECIES DELIMITATION

A Process-Oriented View on Species Delimitation

Perhaps most importantly, genomic data provide a comprehensive evolutionary perspective on species delimitation. Genomic data can be used to infer phylogenetic relationships among species in a clade, to infer the demographic history of species through time (Flouri et al. 2023, Li & Durbin 2011), and to identify instances of hybridization and introgression among closely related species (Hibbins & Hahn 2022). In combination, these methods can provide useful information on the evolutionary processes that drive population divergence and speciation, enabling both the delimitation of species and an understanding of their formation. This process-oriented approach to species delimitation can help reunite the fields of taxonomy, systematics, and speciation biology (Fujita et al. 2012, Smith & Carstens 2020).

A More Accurate Window into Species Boundaries

Genomic data can provide more accurate insights into species boundaries, particularly when other data types are inconclusive or misleading. For example, although approximately 70% of species delimited using morphological data appear to be robust (Rieseberg et al. 2006), morphological delimitations can be inaccurate, particularly when morphological features evolve multiple times independently or when morphology evolves slowly or subtly (Bickford et al. 2007). In such cases, genomic data can help accurately resolve species boundaries.

Further, relative to single-locus data (e.g., DNA barcoding), genomic datasets provide a more accurate perspective on genetic divergence and species delimitation. Genealogical discordance—or when different genes paint different evolutionary stories—is to be expected given the stochastic nature of the coalescence process, introgression, and selection (Degnan & Rosenberg 2009, Dupuis et al. 2012). Thus, relying on single genes or even a handful of genes provides an incomplete view of a species history. This is especially true in the case of organelle genomes, which are thought to be under strong selection due to their pivotal role in organismal physiology and particularly susceptible to introgression due to their inheritance patterns (Petit & Excoffier 2009, Toews & Brelsford 2012). However, because genomic data are sampled across independent regions of the genome, each with its own coalescent history, genomic data can provide a richer, more accurate picture of the history and divergence of species relative to smaller datasets.

An Efficient Way to Generate Homologous Datasets

Because of the deep homology in genome content across species, collecting comparable genomic datasets across clades is relatively straightforward. Further, for many taxa, genomic data are becoming increasingly accessible and affordable to collect, especially relative to morphological data, which can require significant effort when measuring samples widespread across a species' geographic range. (Analyzing data—in terms of both computing power and human expertise—can remain expensive.) These efficiencies make genomic species delimitation an attractive option for explorations of species boundaries.

CHALLENGES AND OPPORTUNITIES OF GENOMIC SPECIES DELIMITATION

Genomic Divergence Is Just a Proxy

While genomic data can provide a quantitative perspective on genetic divergence between species, this is simply a proxy for the processes of species formation, i.e., ecological differentiation and reproductive isolation. Although genetic divergence typically scales with ecological differentiation and reproductive isolation (Coyne & Orr 1989, Funk et al. 2006), this scaling is imperfect, and relying solely on genomic data can result in missing nuance.

As of yet, most genomic species-delimitation studies leverage just one aspect of genomic data: variation in nucleotide sequences. However, interrogating the genome in different ways could introduce more subtlety into characterizations of species boundaries. First, other types of genomic data might integrate more ecological context into genomic species delimitation. For example, in addition to delimiting species using genomes from host species, the genomes of associated symbionts, like microbiomes, parasites, and *Wolbachia*, could be included (Oury et al.

2023). Because many of these symbionts have tight affinities with their hosts, understanding their biology can help reveal the unique biology of the putative species. Alternatively, other types of genomic data such as epigenetic modifications, gene expression levels, and gene–gene interactions can provide insight into morphology, physiology, and other phenotypes relevant to understanding ecological differentiation and reproductive isolation (Pease et al. 2016, Schiffman & Ralph 2022).

Second, instead of focusing on the genome as a neutral marker, genotype and phenotype can be linked to understand how species-specific differences and reproductive barriers are encoded in the genome. While most tractable in recently diverged species, approaches like quantitative trait locus mapping can uncover the genetic architecture of a key species trait—such as differences in flower color that promote reproductive isolation (Schemske & Bradshaw 1999, Wessinger et al. 2023). Comparing genetic divergence across trait-based loci relative to background levels of divergence can help integrate the role of trait divergence into genomic species delimitation (**Figure 4**). As seen in the *Ruminococcus* bacteria, such an approach could be particularly useful in groups that recently diverged or are undergoing gene flow (Cadena & Zapata 2021, Funk et al. 2021), because it can allow focused examination of those genomic regions that make species distinct despite prevailing genomic similarity.

Third, advancements in long-read sequencing technology have renewed recognition of a well-known observation: Species often differ significantly in both broadscale genome structure, such as variation in chromosome number, and structural features, such as inversions and duplications

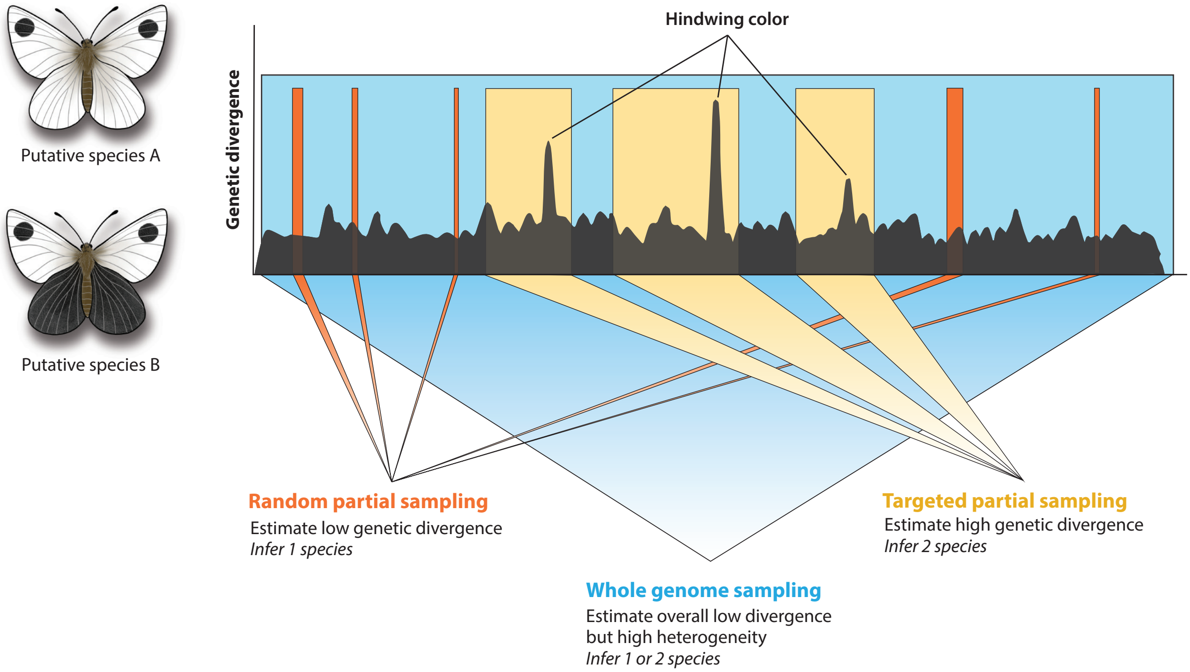


Figure 4

Heterogeneous genome divergence can affect estimates of species limits. In the case of recently diverged, phenotypically distinct species, random partial sampling throughout the genome results in low estimates of genetic divergence, suggesting the two species should be collapsed. If instead we sequence those parts of the genome that underpin phenotypic differences, we infer high genetic divergence, suggesting the two species are distinct. However, if we sequence the whole genome, we uncover high levels of heterogeneity and overall low divergence, and the status of these two species remains ambiguous.

(Berdan et al. 2024, Bush et al. 1977). Both theory and empirical data show that such structural variation can impact evolution more than variation in nucleotide sequences. Effective recombination rates are often reduced between chromosomes that vary structurally, leading loci to be trapped together and increasing linkage disequilibrium. When a recombination block contains multiple genes influencing multiple traits, then phenotypes at these traits can diverge in concert rapidly (Kirkpatrick & Barton 2006, Todesco et al. 2020). Low recombination rates can also reduce gene flow between species (Rieseberg 2001), hastening divergence. Thus, as seen in redpoll finches, considering structural variation could provide remarkable explanatory power for species delimitation that might be missed if just looking at variation in nucleotide sequences (Funk et al. 2021, Sproul et al. 2020).

Ultimately, perhaps the most promising path forward is to expand genomic species delimitation beyond genomic data. While genomic data have become increasingly affordable and accessible to collect, many types of phenotypic and ecological data remain time-consuming to gather. New approaches like community science databases (Hantak et al. 2022) and high-throughput phenotyping methods (Weeks et al. 2023) could alleviate this problem by building phenotypic datasets that scale across individuals as efficiently as genomic methods do. Genomic and phenotypic data would also benefit from a unified analytical framework. Most existing species delimitation methods are limited to modeling genetic data, often under the assumption that these data evolve neutrally. In contrast, nongenetic traits—such as morphological features, life-history characteristics, physiological adaptations, or reproductive barriers—are typically analyzed separately. Further, these traits are unlikely to follow neutral evolutionary patterns. Researchers are then left to synthesize these different datasets on their own. Machine-learning approaches offer one avenue for synthesis (Karbstein et al. 2024, Pyron 2023), but if researchers want a process-oriented model that incorporates neutral genetic loci, adaptive phenotypic traits, and geography, there are none available (but see Solís-Lemus et al. 2015). Such a model could help resolve conflict across datasets and could enable species delimitation in systems that are lacking genomic data, including extinct, fossilized species.

Species Criteria Will Always Be a Problem

While empirical data have solved some challenges in species delimitation, many of the eternal debates on what defines a species persist (Maddison & Whitton 2023, Pigliucci 2003). Delimiting species can be as hard as identifying the criteria that define species (De Queiroz 2007), and genomic data have not (yet) resolved these recalcitrant philosophical questions.

For example, lineage divergence can be a slow and continuous process, but how much divergence is necessary before a lineage is recognized as a distinct and robust species? Genomic data can accurately and precisely characterize some of the demographic processes under which a lineage has evolved, including how long ago the lineage split from its ancestor and the extent of gene flow the lineage experienced throughout its divergence. Understanding this history can help determine whether a lineage meets a threshold to be considered a species, but given the heterogeneity of species formation, thresholds are likely to vary across different taxa. When divergence varies across axes—such that species cross the threshold for phenotypic but not genomic divergence (**Figure 1**)—resolving species boundaries can become even more complicated.

Further, how should asexual species, for which species cohesion cannot be simply maintained by interbreeding, be delimited? One potential solution here—afforded by new genomics approaches—is to be agnostic about species definitions and instead characterize the metagenomic diversity in a given community of asexual species. By characterizing which gene families are present, and in what copy number and diversity, these approaches move from understanding diversity by counting the number of species to characterizing the functional repertoire of a community. While this approach is promising, it ultimately leaves the species-level units undefined, which could hinder future taxonomic studies. Alternatively, as illustrated by the *Ruminococcus* case studies, ecologically relevant variation in the genome under selection can be characterized (Arevalo et al. 2019) to understand how natural selection maintains species cohesion in asexual organisms (Barraclough 2019, Shapiro & Polz 2015).

Genome Heterogeneity Complicates the Picture

Genomes are inherently heterogeneous because the sources of variation—de novo mutations and recombination—act heterogeneously across the genome. For example, recombination is lower than average in genomic regions like centromeres and sex chromosomes, and across structural variants (Peñalba & Wolf 2020). The forces that act upon this variation—drift, selection, and introgression—also vary relative to the effective population size and fitness effects of a genomic region. Because of this, patterns of genomic divergence are heterogeneous across the genome, as has been captured across multiple verbal models in the speciation literature including genome congealing, the genic view of speciation, barrier loci, and islands of speciation (Turner et al. 2005, Wu 2001).

Heterogeneity across the genome is likely the norm, especially given the ubiquity of introgression across the tree of life (Dagilis et al. 2022, Mallet 2005). This heterogeneity can complicate species delimitation, however, because levels of divergence can vary dramatically depending on which subsample of the genome is considered (**Figure 4**). Take a hypothetical example of two recently diverged lineages that differ at just a few phenotypic traits: The loci underpinning these lineage-specific traits are likely to be under strong divergent selection and reduced introgression. If these loci are assayed, genetic divergence is high, suggesting that these two lineages are different species. Conversely, using genes that are subject to neutral processes would result in lower estimates of genetic divergence, instead suggesting that these lineages are acting as a single species.

While this heterogeneity can be confusing, variance in loci divergence reflects the interactions between selection, introgression, and recombination (Burri 2017, Cruickshank & Hahn 2014, Stankowski et al. 2019). Therefore, heterogeneity across the genome can further reveal the processes that structure lineage divergence, providing additional context to delimit species. Moving forward, this heterogeneity can be used to gain another perspective on a lineage's progress to speciation. For example, new methods could explicitly model heterogeneity and account for its extent as a parameter in evaluating speciation progress. Additionally, visualizing this heterogeneity—i.e., by modifying Manhattan plots to plot species delimitation metrics such as the posterior probability of speciation (from programs like BPP)—could also help better evaluate species status.

Geography Remains Confounding

Most species—particularly those spanning broad geographic areas—exhibit genetic structure within their range, as seen in the *Physalia* case study. Thus, determining whether a recognized species consists of multiple, unrecognized species requires broad sampling throughout the range, especially to distinguish between clinal versus disjunct genetic structure (Chambers et al. 2025, Mason et al. 2020). Genomic data partially address this issue because sequencing across the genome allows population-wide dynamics to be captured by a single individual (Li & Durbin 2011), thus reducing the total number of samples that need to be collected. However, the number of localities that need to be sampled remains the same, and sampling should ideally include either holotypes or topotypes to facilitate taxonomic revisions. Ultimately, this challenge highlights the central importance of field-based collections and the critical role that natural history museums play in systematics research (Holmes et al. 2016). These natural history collections allow researchers to expand their geographic sampling by obtaining samples from genetic resource collections or by collecting genomic data from specimens such as toe pads from mammals and birds and pressed leaf material from plants (Bi et al. 2013).

Translating Revised Species Boundaries into Taxonomy Can Be Slow

While collecting genomic data is becoming a popular choice for many taxa, potentially accelerating species delimitation, implementing taxonomic changes remains a bottleneck in most groups. In some prokaryotic and single-celled eukaryote groups, scientific norms have adjusted, allowing researchers to write shorter, more efficient species descriptions that can keep pace with the rate of data acquisition (Renner 2016). However, in most other groups, formally describing newly delimited species remains time-consuming, requires specialized knowledge in natural history and taxonomy, and is seen as low-impact science. Given these costs, it is perhaps unsurprising that many studies are reluctant to propose formal taxonomic revisions until they can collect additional data corroborating the genomic species delimitation. Ultimately, only ~36% of the studies included in our review of genomic species delimitation revised taxonomy.

That said, the use of genomic data in systematics could also serve to infuse new interest and importance into taxonomy and species delimitation. Taxonomy is sometimes seen as an outdated field, but the integration of genomics and machine learning into taxonomy reenvisions the field as a big-data science. Integrating genomics into how students are trained in species delimitation and taxonomy and sharing with the public how genomic data can help count the number of species on Earth could serve to reinvigorate the field of taxonomy (Gorneau et al. 2022).

Species Delimitation Approaches Can Return Discordant Results

Perhaps it is unsurprising when genetic and phenotypic perspectives on species boundaries conflict given the differences in how these characters evolve (**Figure 1**). However, different analytical approaches can return discordant results when fed the exact same data. This variability arises from differences in the underlying modeling frameworks of the methods (Hausdorf 2025, Rannala 2015). For example, population clustering approaches try to maximize linkage and Hardy–Weinberg equilibrium, whereas MSC approaches evaluate whether the coalescent patterns of the loci are better described by a single or multiple populations. Such discordance can stymie researchers from formalizing taxonomic revisions, limiting the applicability of species-delimitation studies (Carstens et al. 2013).

We see two potential paths forward. First, rather than seeking consensus among multiple methods that all make poor assumptions, we suggest applying a narrow set of complementary analyses for species delimitation, carefully picking approaches whose assumptions align with the dataset (Rannala 2015). For example, instead of using three MSC approaches on the same dataset, it would likely be more fruitful to apply a single well-justified MSC framework and add an approach that accommodates geospatial patterns, like evaluating isolation-by-distance patterns within and across putative species (Hausdorf 2025). Second, machine learning and other forms of artificial intelligence can integrate multiple datasets for composite species delimitation (Derkarabetian et al. 2019, Karbstein et al. 2024, Pyron 2023). The speed of these approaches could also serve to reduce the taxonomic impediment. However, the machinery underpinning these approaches is typically quite opaque, unlike traditional methods with clearly defined, evolutionarily informed models. Thus, these approaches are not easily interpretable in light of biological processes, unless users provide a post hoc analysis or description of the results recovered.

OUTSTANDING ISSUES

Reproducibility of Genomic Data

Taxonomy is an iterative process that requires species limits to be retested as we acquire new samples and data. Building on existing genomic datasets can be challenging, however. Data processing and filtering can have a marked influence on inference (Hemstrom et al. 2024, Shafer et al. 2017), so existing datasets can be expanded only when associated with a reproducible workflow. Further, particularly for reduced-representation approaches, batch effects can reduce comparability across datasets. In addition, genomic datasets are not always associated with important metadata like geographic coordinates (Toczydlowski et al. 2021). We can address these challenges by strengthening community norms around the sharing of genomic data, their metadata, and their associated analysis pipelines, thus allowing future researchers to revisit the species delimitations of today.

Equity Issues in Genomic Species Delimitation

While the costs of sequencing have dropped six orders of magnitude since 2000, collecting genomic data remains prohibitively expensive and inaccessible for many researchers. Further, genomic datasets require access to high-speed networks to download and share the data, multicore computers with high memory footprints and reliable power sources to process the data, and redundant hard drive storage to preserve the data long-term (but see Handika & Esselstyn 2024). These resources are not universally available, nor is the training to collect and analyze these data. Expecting or requiring genomic data for species-delimitation studies could inadvertently exacerbate existing inequities in the field and further bias taxonomic shortfalls (Linck & Cadena 2024). Indeed, the regions predicted to have the greatest shortfalls (Freeman & Pennell 2021) are some of the same regions with less developed research infrastructure (**Figure 3*a***). We thus caution against setting genomic data as the gold standard for the field; many of the most pressing, practical concerns of species delimitation can be effectively and efficiently addressed using smaller datasets (Bertola et al. 2024).

CONCLUSION

While genomic data are rapidly transforming species delimitation by offering unprecedented resolution and analytical power, it is increasingly clear that genomics alone may not be the holy grail. The power of genomics lies not in the data or the methods per se but in their ability to inform fundamental biological questions about speciation, adaptation, ecological differentiation, and evolutionary history. Without anchoring genomic species delimitation analyses in the biological reality of organisms, we risk generating species boundaries that are analytically elegant but biologically irrelevant. Thus, as species delimitation becomes an increasingly quantitative big-data science, we must remain deeply rooted in the biology of the organisms. Which individuals should be sampled, what data should be collected, which conceptual framework and analytical approach should be applied, how results should be interpreted (**Figure 2**, **Table 1**, **Supplemental Table 1**): The answer to each of these questions depends on the study species, its unique biology, and the landscape in which it is found.

Supplemental Material >

Ultimately, we see species delimitation as just the first step. If genomic species delimitation results remain in an abstract space of statistical entities without clear ties to recognizable biological units, they risk being disconnected from biological research, conservation efforts, and policy decisions. Fortunately, genomic species delimitation has made it easier to apply our delimitations to taxonomic and evolutionary questions. Even if many challenges and open questions persist (see the Future Issues), the progress afforded by genomic species delimitation is an opportunity to integrate a genomic perspective with understandings of species from other perspectives—including their natural history, phenotypic divergence, geographic distributions, and reproductive barriers. As we move forward, the greatest promise of genomics then lies not just in refining species boundaries but in enabling a more integrative, biology-driven approach to studying life's diversity.

FUTURE ISSUES

1. What other types of genomic data could we collect to aid in genomic species delimitation?
2. How can genomic species delimitation methods include other sources of data—e.g., morphological differentiation, reproductive barriers—to create an integrative delimitation approach?
3. How can the extent of heterogeneity in genomic divergence be used as a metric for capturing species divergence, and thus, in delimiting species?
4. How can genomic species delimitation help address the taxonomic impediment?
5. How can genomic species delimitation help unite the fields of taxonomy, systematics, and speciation biology?
6. How can technological advances like artificial intelligence help delimit species more efficiently and effectively?

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